SUMMARY

- A disturbance which carries energy and momentum from one point in space to another point in space without the transfer of medium is known as a wave.
- The waves which require medium for their propagation are known as mechanical waves.
- The waves which do not require medium for their propagation are known as non-mechanical waves.
- For a transverse wave, the vibration of particles in a medium is perpendicular to the direction of propagation of the wave.
- For a longitudinal wave, the vibration of particles in a medium is parallel to the direction of propagation of the wave.
- Elasticity and inertia are necessary properties of the medium for wave propagation.
- Waves formed in still water (ripples) are transverse and wave formed due to vibration of tuning fork is longitudinal.
- The distance between two consecutive crests or troughs is known as wavelength, λ.
- The number of waves which crossed a point per second is known as frequency, *f*.
- The time taken by one wave to cross a point is known as time period, *T*.
- velocity of the wave is $v = \lambda f$.
- Frequency is source dependent and wave velocity is medium dependent.
- The velocity of a transverse wave produce in a stretched string depends on tension in the string and mass per unit length. It does not depend on shape of the wave form.
- Velocity of transverse wave on a string is $v = \sqrt{\frac{T}{\mu}}\, ms^{-1}$.
- Velocity of longitudinal wave in an elastic medium is $v = \sqrt{\frac{E}{\rho}}\, ms^{-1}$.
- The minimum distance from a sound reflecting wall to hear an echo at 20°C is 17.2 meters.
- The wave equation is $\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}$ in one dimension.
- Wave number is given by $k = \frac{2\pi}{\lambda} rad\, m^{-1}$.
- During interference the resultant intensity is $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\varphi$, where the intensity is square of the amplitude $I = A^2$.

  For constructive interference, $I_{\text{maximum}} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2 = \left(A_1 + A_2\right)^2$.

  For destructive interference, $I_{\text{minimum}} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2 = (A_1 - A_2)^2$.
- When we superimpose two or more waves with slightly different frequencies then a sound of periodically varying amplitude at a point is observed. This phenomenon is known as beats. The number of amplitude maxima per second is called beat frequency.

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